

day-49-heygen-exact-scope-proof

Day 49 - HeyGen Exact-Scope Proof

Date: 2026-05-10

Lane: HeyGen Exact-Scope Proof

Status: CLOSED AS BLOCKED

Blocker class: BACKEND_MISSING

Objective

Implement the exact-scope HeyGen generation proof path behind Bridge approval without running HeyGen, invoking Zapier writes, or creating video output.

Result

- Added protected endpoint: 'POST /api/bridge/heygen/exact-scope-proof'.
- The endpoint uses Day 48 schema readiness first.
- If HeyGen schema is not visible, the route returns a safe blocker and does not create an approval request.
- If schema is visible and payload is invalid, the route returns missing fields and does not create an approval request.
- If schema is visible and payload is valid, the route creates a pending Bridge approval request scoped to 'heygen.generate'.
- The approval scope stores a payload hash and field names only, not raw script/body values.
- The route always keeps generation disabled.

Current Runtime Truth

Current live runtime cannot see Zapier MCP/HeyGen schema:

- 'status: 503'
- 'mode: heygen_exact_scope_blocked'
- 'required_scope: heygen.generate'
- 'blocker: MCP server zapier is not present in the Claude MCP configuration.'
- 'approval_request_created: false'
- 'accepted_for_execution: false'
- 'execution_enabled: false'
- 'writes_enabled: false'
- 'no_heygen_generation: true'
- 'no_zapier_writes: true'

Exact blocker classification for the day: 'BACKEND_MISSING'.

Files Changed

- 'src/app/api/bridge/heygen/exact-scope-proof/route.ts'
- 'src/app/api/bridge/heygen/exact-scope-proof/route.test.ts'
- 'scripts/check-heygen-exact-scope-proof.mjs'
- 'runtime/day-49-heygen-exact-scope-proof.json'
- 'runtime/day-49-heygen-schema-readiness-proof.json'
- 'runtime/day-49-route-rendering-proof.json'
- 'runtime/day-49-heygen-exact-scope-proof.md'
- 'runtime/day-49-heygen-exact-scope-proof.pdf'

API Behavior

Unauthenticated:

- 'POST /api/bridge/heygen/exact-scope-proof' returns '401'.

Authenticated with current backend missing:

- Returns '503'.

- No approval request is created.
- No execution or write path is enabled.

Authenticated with schema visible and valid payload:

- Creates a pending Bridge approval request for 'heygen.generate'.
- The request uses:
 - 'connector: zapier.heygen'
 - 'action: heygen.generate'
 - 'riskLevel: high'
 - 'protectedCategory: external_automation'
 - 'targetKey: <visible HeyGen tool name>'
- The route still returns 'accepted_for_execution: false'.

Runtime Proof

Commands:

```
""bash
node scripts/check-heygen-exact-scope-proof.mjs http://127.0.0.1:3337
node scripts/check-heygen-schema-readiness.mjs http://127.0.0.1:3337
node scripts/check-mission-control-route-rendering.mjs http://127.0.0.1:3337
""
```

Results:

- HeyGen exact-scope proof smoke: PASS, 'checked: 2', 'failures: []'
- HeyGen schema readiness smoke: PASS, 'checked: 3', 'failures: []'
- Route rendering smoke: PASS, '46' routes and '8' designer pages
- '/login': '200'

Proof artifacts:

- 'runtime/day-49-heygen-exact-scope-proof.json'
- 'runtime/day-49-heygen-schema-readiness-proof.json'
- 'runtime/day-49-route-rendering-proof.json'

Validation

- TDD red: endpoint test first failed on missing route.
- Focused test: PASS, '1' file / '4' tests.
- 'pnpm run typecheck': PASS
- 'pnpm run build': PASS
- 'pnpm test': PASS, '175' files / '1375' tests
- 'git diff --check': PASS
- 'node scripts/check-protected-file-invariants.mjs': PASS
- '.env' diff check: clean
- Staged secret scan: run immediately before commit.

Runtime

- Mission Control standalone runtime restarted on '127.0.0.1:3337'.
- Runtime PID after restart: '12295'.
- No public local exposure added.
- No '.env' change.

Safety

- No HeyGen generation.
- No Zapier writes.
- No external writes.
- No Bridge bypass.
- No approval request is created while schema is unavailable.
- Raw payload/script text is not returned by the endpoint or proof script.

- No fake sent/done/live state.
- No secrets printed.
- No auth weakening.

Commit / Push

- Commit: recorded after this report is committed.
- Expected message: 'feat(heygen): gate exact-scope proof approval'
- Push target: 'origin/to-knowledge-mc'

Rollback

```
""bash
git revert <day49_commit_sha>
""
```

Remaining Blocker

'BACKEND_MISSING'

HeyGen exact-scope proof can move forward only after:

- Zapier MCP/backend is configured in the runtime.
- HeyGen appears in the Zapier MCP tool inventory.
- A valid payload is submitted through the schema-readiness contract.
- Owner approves an exact 'heygen.generate' Bridge request before any generation runner is allowed.